University of Cape Town
Department of Computer Science

Computer Science CSC3003S
Class Test 1, 2023

Answer all questions.
Approximate marks per question are shown in brackets.

Time: 40 minutes **Marks: 35**

Section 1. ASD1: Project Management [23 Marks]

Refer to the project brief below and answer the questions relating to the software development methods:

As the founder of a new startup, you are developing a platform that will screen job seekers for software engineering skills and match them with prospective employers. The screening process entails review of submitted documents such as the CV and an online test. Prospective employers may search through applicants that have passed the screening, and the system would ideally also suggest applicants with matching skills. There are a few potential and incumbent competitors, so you have already gone to market with manual screening and matching, but your platform is intended to streamline and automate your operations. You have a small team: three experienced software developers, a human resources person experienced with software recruiting, and a project manager, all of whom have worked together previously with you in another startup. You currently have funding to operate for about one year.

Question 1 [8 Marks]

a. You will take an agile approach for this project rather than waterfall. Use details from the brief above to justify your decision. [3]

In agile – scope is not fixed, while time and cost are. Scope is adjusted and prioritized so that the most important bits are completed first and additional features are only completed if time and budget are available. While in waterfall

+1 for capturing agile triangle (this may duplicate one of the points below) but for this point the student should note time/scope/cost together

*Any **two** of the following (1 each) or other reasonable arguments for agile:*

- *TIME "potential and incumbent competitors" **Need for frequent working releases***
- *SCOPE - in a **startup**, it is necessary to respond to change*
- *SCOPE - iterative approach allows for incremental delivery*
- *SCOPE - there is the potential to engage with existing customers to learn*

b. Using this project as an example, explain these terms in the context of Scrum projects: Product Backlog, Sprint Backlog, minimal viable product. [3]

Product backlog: the list of high-level tasks to be done for the overall product, ordered by priority. In this case, this would include screening submitted documents, online test, employer search, listing of candidates, accepting applicants, etc [1]

Sprint backlog: a subset of the product backlog, where the selected tasks from the product backlog are expanded into more granular tasks for the sprint. In this case this might be the online test, broken down into specific assessments, scoring, usability assessments, etc [1]

Minimum viable product: a subset of features that are necessary for a releasable product. Here, there's already a manual process which acts as an MVP, so any one feature (screening, online test, listing of candidates for searching) could serve as an MVP as long as it is complete (e.g. tests) and presentable; [1]

c. Consider the Sprint Review Meeting that takes place at the end of each sprint in Scrum project management. What agile principle does this embody? [2]

The above answer is correct for the Sprint Retrospective meeting, not the sprint review meeting. If they return for remarking, then this answer should be remarked with this guide.

Acceptable Principles (2pts for any one of the below) – doesn't need exact wording but should capture the gist

- Satisfy the customer through early and continuous delivery of working software
- The most efficient and effective means of communication is face to face conversation
- Business people and developers should work together
- Deliver working software frequently
- Working software is the best measure of progress
- Welcome changing requirements

The value this embodies (for partial credit, either one):

- Working software over Comprehensive Documentation
- Customer Collaboration over Contract Negotiation
- Responding to Change over following a plan
- Individuals and interactions over processes and tools.

Effectively – if they named a value, they should get one point (arguably they all apply), and they named one of the above principles they will get two points for identifying a relevant principle.

Question 2 [6 Marks]

a. Your team has not used a test-driven development (TDD) approach before, but you believe that this is the right approach for the project. Write a short memo that explains what TDD is and gives at least two reasons to motivates its use in your team. [3]

TDD is test-first-development + refactoring, which means that you write tests before writing functionality, and you use these tests to verify a working system when you clean up the code. [1] (can give partial credit for saying only tfd)

Test first development is great because it ensures that all the testing framework is thought through alongside coding, but it can lead to messy code due to organic growth.

2 points for any of the following or other logical reasons for adopting TDD

Writing the tests before code clarifies the requirements [1]

Refactoring counters software entropy [1]

TFD gives assurance that refactoring hasn't broken prior functionality [1]

b. Explain what user acceptance testing is and explain how you would integrate it into your project plan. Is this a functional or non-functional test? [3]

User acceptance testing consist of non-functional tests [1] run to ensure that this is what the end-users or customers want. [1] In this case the startup is developing directly for consumers – software developers and potential employers, so UAT would be evaluated through alpha and beta testing of the product at each release. Data points may include software reviews and downloads, but could also include focus groups or other qualitative approaches to working with end-users [1 for saying getting customer input before and during each release]

Question 3 [3 Marks]

Define **critical path** in software engineering. Give an example of a potential software dependency in this project and use this to explain the importance of understanding critical paths. [3]

The critical path is the tasks in the project where any additional time required to complete a task will delay the whole project. [1 for definition]

It can be used to estimate how long the project might take, and how to prioritize and allocate resources to tasks to eliminate delays due to dependencies [1 for importance]

Examples might include needing a sign-on mechanism to authenticate potential employers for using the system, or needing to have the assessment and processing of candidates done in order to develop the matching algorithms [1 for any reasonable example]

Section 2. ASD2: Software Architecture [13 Marks]

Question 4 [1 Marks]

Give the statement number of the correct statement below: [1]

1. Good quality software has high coupling and high cohesion
2. Good quality software has high coupling and low cohesion
3. Good quality software has low coupling and low cohesion
4. Good quality software has low coupling and high cohesion

4

Question 5 [2 Marks]

Performance testing should not only test system performance under expected normal loads. What 2 other types of testing are needed? Just explain each in 1 short sentence (no need to name them). [2]

Any 2 of:

sudden spike in demand,

higher & higher loads (test breaking point),

running longer & longer under normal load (longer duration)

larger & larger volumes of data (scalable volume)

Question 6 [3 Marks]

Briefly explain the following:

1. Code churn [1]
2. Fault tolerance [1]
3. Polymorphism [1]

1. code often edited

2. ability to continue running when some things go wrong/fail

3. ability to use the same method on different types of object

Question 7 [2 Marks]

Suppose Rectangle and Oval are defined as 2 subclasses of Shape. Which SOLID property would be broken if Square was then defined as a subclass of Rectangle, and why? [2]

Liskov Substitution Principle ✓

As not substitutable/different properties/different behaviour because rectangle has width & height, whereas square only has one/same ✓

or

Open Closed Principle ✓

As Square must *change* height & length to one/same attribute, *not extend* Rectangle ✓

Question 8 [1 Marks]

A Bookshop class has a List of Book objects sold. The shop wants to sell Video and Audio objects now, which are similar to Book objects but not exactly the same. The Bookshop class has to be changed, not extended, because it didn't follow the Open Closed Principle. What mistake was made in Bookshop, and what should have been done instead? [1]

Instead of using the class Book in the List, they should use an interface

Question 9 [4 Marks]

Answer the questions that follow about the code below that is using the Factory pattern:

1. What word should replace ???? [1]
2. What Java statement is typically found in the missing part of getProduct? (answer is one reserved word e.g. WHILE) [1]
3. Give the 2 missing statements of the main( ) method to show how Factory can be used to make a product and store it in variable myPen. [2]

1. IProduct

2. either CASE or IF

3. Factory something = new Factory( ); ✓

IProduct myPen = something.getProduct("pen"); ✓

```
public interface IProduct {
        String getName( );
}
public class Factory {
          public  ????     product;                              // answer to (1)
          public  ????     getProduct( String productType) {     // answer to (1)
                    // method code not shown here                // answer to (2)
                    return whatever;
           }
}
public class App {
        public static void main(String[ ] args) {
              // declare variable myPen and store a new "pen" productType there: (3)
        }
}
```

-END-